How to read these notes

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This part contains 4 notes, from `AGENTS.md` through `tools/release/how-to-read.md`.

`AGENTS.md`

Agent rules for Graal_analysis

Owner: Squirrel (GitHub `squirelworks`). Merges and tags: Chiaki only. Agents open PRs; they never merge, never push to `main`, never create a tag or release.

No real servers, no binary addresses

- **No real server IPs, ports or hostnames** (public Graal servers, the official listserver and login hosts, anyone's LAN machines) in notes, TOML/TSV tables, logs, scripts or screenshots. Use the loopback address, a second loopback address (fake outside interface), the lab ports (14802, 14803, 14804, 14900, 14921–14923), or placeholders `<listserver-host>`, `<login-host>`, `<public-host>`, `<lan-host>`. When a historical host matters, describe it ("the 1999 list host, same as 1.39's") rather than naming it. Do not write dotted IP literals anywhere in the repo.
- **No binary addresses**: no `0x004…`-style offsets, no Ghidra/IDA names (`FUN_…`, `DAT_…`, `LAB_…`), no decompiled C. Findings are written as plain behaviour with a source tag (`[2.05]`, `[bin]`, `[lab]`, `[cap]`, `[srv]`, `[doc]`, `[inf]`).
- The repo is **public**. Treat every commit as published.

Labs

- **Cloud loopback only**: servers you start yourself on the loopback address (or a second loopback address as a fake outside interface), in a container, VM or CI runner.
- **Never use Squirrel's own machines** (their Linux PC, Windows PC, Mac) for labs, captures or builds.
- **No public servers** (no logins, no listserver registration, no captures against them) without Squirrel's explicit OK in writing.
- Never run old Graal tools from the archives outside a sandbox: `ListServer.exe` and several clients phone home. Wine + Xvfb in a container or a qemu guest is fine.
- Every lab note records: client + version, server + commit, what was done, before, after, and log/pcap file names (redacted per the host/IP rules above).

Decompile and pyReborn material stays here

- `Graal_analysis` is the **only** repo where decompile-derived findings, pyReborn study notes, GPL server notes and the reference decoders/scripts live.
- **Never copy any of it into the clean-room Rust repos** (`seihai`, `seihai-gserver`, `seihai-listserver`, `seihai-rc`, `seihai-npcserver`, `seihai-servertools`): not text, not tables, not code, not the Python decoders. Those repos may *link* to a note here; their code is written from their own protocol docs, lab captures and tests.
- Raw decompiles (Ghidra `.c`/`.xml`/`.html`, `.idc`, `.gzf`) never enter any repo. Squirrel holds them (Drive / project files). If a task needs one, ask Squirrel for it and commit only the plain-behaviour result.

- pyReborn (`hosler/pyReborn`, `reborn-protocol`, `pygserver`) is MIT: summarise and cite (`file:line`), do not paste code. Its "reversed official client" citations are hints only; confirm against our own decompile notes or a lab run.
- GPL patches for the lab servers (`tools/lab-servers/*.patch`) stay here and in the GPL fork (`seihai-gserver-v2`, `seihai-gs2lib`), nowhere else.

Git and PRs

- **Verify merge state before you start any task**: `git fetch origin main`, check that dependency PRs are merged and that no open PR already covers the task. If a PR you depend on is still open, do not stack on it; take the next task.
- One change per branch and PR. Commit message: short summary, then why, then what evidence (lab run, static read, or "none").
- Before pushing, run the leak checks below on your diff. A PR that adds a host, address or `FUN_` name is not ready.
- Add the new note to `README.md`'s contents list in the same PR, and close the item in the matching open-points file (strike it and link the new note).
- **Stop and ask Squirrel** (leave the task, go to the next) where a task says "Squirrel decides".
- Leave the built-in guest password as it is on `main`. Do not redact it again.

Leak checks (run from repo root before push)

```
git ls-files | grep -E '\.pyc$'
git grep -nE 'FUN_[0-9a-fA-F]{6}|DAT_[0-9a-fA-F]{6}|\b0x00[4-6][0-9a-fA-F]{5}\b'
git grep -nE '\b([0-9]{1,3}\.){3}[0-9]{1,3}\b' | grep -vE '\b[0-9]\.[0-9]+\.[0-9]+\.[0-9]+\b'
git grep -niE 'graalonline\.com|graal\.in\b|graalreborn\.(com|net)'
```

(The third line skips version strings like `2.2.2.0`; review its hits by eye.)

Task list

See `docs/NEXT.md` for the ordered open work.

Cursor Cloud specific instructions

Python 3.12 is already on the image. The environment install adds pytest for the login user (`python3 -m pip install --user pytest`). That is the only package `.github/workflows/checks.yml` installs. `python3 -m pytest` is the command to use; the `pytest` script also lands on `~/.local/bin`, which login shells already include.

From the repo root:

- `python3 -m pytest tools/lab-servers -q` starts the lab listserver, drives every login shape, and stops it. That test has to be the only gserver connected, so do not leave another listserver or gserver bound to the lab ports.
- Smoke `client-internals/version-profiles/resolve.py` on a profile TOML, and `formats/grv1-decoder/grv1.py` on a level, the same way the workflow does. Both use the standard library only.

Do not start archived Graal clients or Wine here. Those clients phone home, and the client archives are not in this checkout. Lab servers stay on the loopback address.

`docs/NEXT.md`

Next work (ordered)

One change per branch and PR. Verify `origin/main` and the open PR list before starting; do not stack on unmerged PRs. Skip items that say "Squirrel decides" or that need missing Squirrel-only files; take the next task. Binding rules live in `AGENTS.md` and `.cursor/rules/graal-analysis.mdc`.

Done-checks also require: leak check clean, `README.md` contents updated, matching open-point closed.

Tasks

- [x] **1. Repo rules and task list.** Add `AGENTS.md`, `.cursor/rules/graal-analysis.mdc`, and this file. Fix the README opening line to match its Boundary section ("no raw decompiles, code or binary addresses"). *Done in #152.*

- [x] **2. Leak scrub.** One PR per kind: (a) #154 `.pyc`; (b) #153 `FUN_` names;

    1. #156 IPs/LAN/personal machines; (d) #157 official hostnames; follow-up #158 loopback dotted literals → plain words / `<loopback>`. Guest password untouched. *Done when* the four leak greps in `AGENTS.md` return nothing (third line: review hits by eye).

- [x] **3. CI for the checks.** `.github/workflows/checks.yml`: four AGENTS.md leak greps (fail on any hit; IP grep has no loopback whitelist), `pytest tools/lab-servers`, smoke `resolve.py` + synthetic GR-V1 `grv1-decoder`. Fail-demo branch `cursor/ci-leak-checks-fail-demo-b5fd` proved the IP step red. *Done in #159.*

- [ ] **4. pyReborn study into the repo.** Needs Squirrel's `pyreborn-vs-seihai.md` and `parser-probe-main.rs` (project files). Add as `client-internals/pyreborn-study/` with the MIT note; keep `file:line` citations; drop pasted pyReborn code. *Blocked until Squirrel provides the files.*

- [x] **5. Graal3D collision as plain behaviour (unblocks `seihai-g3d` G3D-1).** Per-face step test in `graal3d/collision.md` with tags: radius 5, step height 30, slope (ny > 0 / attach > 0.2), wall distances 3 / −5, probe order. Static PE float read of `Graal3D.exe` (no decompile on Drive; no C committed). Wine lab still needs native `d3drm.dll` for live confirmation. *Done in #164.*

- [x] **6. Graal3D camera (unblocks G3D-2).** Follow distance 220, look height +80, turn ±5°/tick, pitch −0.15 rad, follow rule **[bin]** (`graal3d/camera.md`, #163). Field of view is the Direct3DRM initial field **0.5**: a square window from −0.5 to 0.5 on the front plane at distance 1, opening `2×atan(0.5)` ≈ 53.13° on either axis. No field constant in the client PE **[bin]**; the value is the library default **[doc]** (SDK `SetField` / `SetFront` remarks, and Wine's native-matching `d3drm` viewport test). The degree figure is arithmetic on that plane **[inf]**. Live confirmation of the follow numbers still needs native `d3drm.dll`. The far plane stays open (`graal3d/camera.md`): library initial back plane 100 **[doc]**, a later DelphiX fork calls `SetBack(5000)`, and the February 2000 exe was not re-read for `SetBack`. *FOV pinned in `graal3d/camera.md`.*

- [x] **7. NPC-Control protocol note (unblocks `seihai-rc` stage 5).** Write `protocol/npc-control.md` from `tools/remotecontrol.md`, Squirrel's `npccontrol` material and a loopback NC run against the GServer-v2 lab. *Done in #155.*

- [x] **8. `#n` in `setcharprop` (server brief A3.6).** What original servers did with `#n` in a `setcharprop` value, from lab evidence (GServer13 1999 under Wine on loopback, and the 2.05 client). Add to `npc-library/gs1-reference-205.md` or a short `server-notes/` note. *Answer in `server-notes/setcharprop-hash-n.md` (#160): value `#n` → action player's nick **[srv]**; GServer13 has no `setcharprop` **[bin]**; live GServer-v2+2.05 lab still open.*

- [ ] **9. Damage and file-transfer open points (loopback captures).** Partial close without new Wine (#165): strike + link for bomb/arrow/baddy powers (`damage-sources.md`), CURPOWER + respawn + default-weapon partial (`lab-captures-2026-10-25`), GWEBL001/`PLO_LEVELMODTIME` (`formats/gwebl001-cache.md`), save folders 1.39/2.00 + empty-stream guard (batch k), pre-2.1 settled-name no re-check (batch e `01-request-lists`). Static read of 2.05 rev 2 and 2.22 (`decomp-findings-2026-10-25-e/04-name-index-case-link.md`): file-index order (newer file in 2.22, first hit in 2.05), plain A–Z case fold, and packet 40's link (a horse does not change it). **Still open:** full vs empty login default checks (the set, and whether they share a frame); maps folder in 2.1–2.16 (the client archive still has no build between the 2.1 test builds and 2.22).

- [x] **10. Long handler layouts per build.** PLAYERPROPS (9), OTHERPLPROPS (8), NPCPROPS (3) and LEVELLINK (1) for 2.00, 2.04, 2.05 rev 2, 2.17.1, 2.22 and 2.31, plus UDP ids 0/15/22 for 2.00–2.04 and gen-3 insertion for 2.01–2.04. Tables: `version-profiles/plo-*-layouts.tsv`, `udp-ids-200-204.tsv`, `gen3-insertion-200-205.tsv`. Open questions 1, 3, 4 in batch-c `10-…` struck. *Done in #162.* (2.04 / 2.01–2.03 `.c` not on Drive; those rows are endpoint + batch-c bracketed.)

- [ ] **11. 1998 to 1.41 rules (matrix unknown #1).** Death, hurt and knockback, sword reach and the wall test for 1.39 first, then 1.32 and ~~1st Graal~~. Decompiles plus Wine runs against GServer13 or the 1998 relay on loopback. *1.39 done in #161 / `version-profiles/matrix/rules-139.md`. 1st Graal done in `version-profiles/matrix/rules-1st-graal.md` (static read of the VCL-lossy export; the death-counter increment is not tagged). 1.32 remains: the Decompile folder was listed again on 2026-11-01 (client exports and Accessories) and still has no 1.32 export, so no death, hurt or wall number was added.*

- [ ] **12. Keep Seihai's fixture reader working.** Any PR that edits `version-profiles/*.toml` or `client-internals/version-profiles/*.toml` runs Seihai's `tools/gen_profile_fixture.py` against the edited file. Do not commit the generated fixture to `seihai` from here.

- [ ] **13. Later, Squirrel decides:** a capture of a real 5.x/6.x `Login` session to fill matrix unknown #3. Ask before planning; public servers are off-limits without Squirrel's OK. Proceeded without a capture (`protocol/login-session-known.md`): what the existing notes already say a Login server sends, and what is still unobserved. The look itself stays open.

README.md

Graal Analysis

Cleaned notes on Graal formats, protocol, NPC scripting, and level design. No raw decompiles, code or binary addresses.

Contents

- `release/0.0.1/` — document PDFs for version 0.0.1, grouped along this contents list, with appendices. Chiaki tags `v0.0.1`. The notes viewer is part of the source tree at `tools/navigator/`.
- `AGENTS.md`, `.cursor/rules/graal-analysis.mdc`, and `docs/NEXT.md` — binding agent rules (no real hosts, no binary addresses, loopback labs only, decompile material stays here) and the ordered open task list.
- `.github/workflows/checks.yml` — CI leak greps (no tracked `.pyc`, no `FUN_`/`DAT_`/`0x004`…, no dotted IP literals, no official hostnames), `pytest tools/lab-servers`, `pytest tools/navigator`, `pytest tools/classic-tools`, `pytest tools/level-convert`, `pytest tools/generators-222`, and `pytest formats/grv1-decoder tools/level-tools-gaps`, plus smoke runs of `resolve.py` and `formats/grv1-decoder`.
- `npc-library/` — 3,172 unique NPC scripts scraped from GR-V1 `.graal` levels, plus tile-neighbor patterns. See also `gs1-commands.md` for the cleaned command catalog and `gs1-reference-205.md` for the 2.05 command, function, and constant tables. Archival notes on a prior larger documented extract are in `npc-library/archival-scripting-notes.md`.
- `formats/` — GLEVNW01, GANI, GRMAP001, tiletypes, board and system notes, and save games (`graalsav.md`). See also `tile-walk.md`, `sign-text.md`, `gobjset01.md`, `level-binary.md` (GR-V1/Z3-V1), and `gmap.md`.
- `protocol/` — packet layouts, login/walk framing, version tokens, capture notes. See also `session-opcodes-cleaned.md`, `codec-and-framing.md`, `packet-ids.md`, `player-props.md`, `file-transfer.md`, `chat-pm-rc.md`, `npc-control.md` (NC login `NCL11012` and NPC/class/weapon packet layouts for `seihai-rc`), `listserver.md` (with `public-listserver-2026-10-25.md`, the first observed public replies), `udp-p2p.md`, `v3x-clients.md` (Graal 3.x Linux/Mac clients), `v4x-clients.md` (Graal 4.x Windows/Mac clients), `v5x-clients.md` (Graal 5.x Windows clients), `v6x-clients.md` (Graal 6.x Linux, Mac and Windows clients, incl. the 6.11 server-defined protocol), `v5v6-windows-unpacked.md` (the packed 5.12–6.03 Windows builds, unpacked and run: tokens, logins, id tables), `v5v6-lab-play.md` (those builds played on a lab GServer: the "Login" server mode that hides characters, 5.12's cipher step count, the session on the wire, push/lift/sword/hide/pot/stone timings and the chat text look), `login-session-known.md` (what the existing notes already say a 5.x/6.x Login server sends, and what is still unobserved; no public capture), `relay-1998.md` (the 1998 text relay), and `gserv-1999.md` (GSERV007–018 logins, gen 2).
- `client-internals/` — version-id map, offline-play notes, Zelda Online 1998 precursor analysis. See also `config-and-input.md`, `rendering-observations.md`, `offline-training.md`, `client-assets.md`, `foot-test.md`, `version-tokens.md`, `client-architecture-205.md`, `des-and-hidden-strings.md` (hidden hosts, 2.22 saved passwords), `newworld-betas.md` (the two NewWorld beta clients), and `live-trace-early.md` (1.1.9 to 1.32 run under Wine against their bundled servers).
- `graal3d/` — the February 2000 Graal3D client and its GServer: protocol (TCP + peer-to-peer UDP), props, accounts, `.x` levels, how it relates to the 2D GServer, client movement/camera/chat (`camera.md`: follow 220, look height +80, field 0.5 / about 53.13°; far plane still open), per-face collision (`collision.md`: radius 5, step height 30, slope and wall rules), a live Wine lab and a reference server. Also `torque-engine-4x.md`: the Torque-based 3D plug-in of the 4.x clients and its UDP protocol on port + 1.

- `torque/` — the Torque Game Engine side of the 3.x–5.12 clients: engine version and lineage, how a client joins a 3D server (Graal TCP + Torque UDP on port+1), the Torque handshake per version (live-captured from 5.12), the `.dso` v33 TorqueScript format with a decompiler, the 5.12 3D object model, 3D formats, and what a Seihai Torque client would need. `torque/wire-by-build.md` compares the wire per build (3.04, 3.153, 4.04r2, 4.20r6, 5.12): game string and protocol, the class table, the networked class sets, the 5.12 game packets and the `GameObject` update.
- `tools/` — cleaned notes on GraalEditor, GraalShop, generators, gif2mng, RemoteControl, and `lab-setup.md` (the offline capture lab). `tools/lab-servers/` holds our lab listserver. `tools/classic-tools/` is the 1999–2001 listserver, stat pages, character designers, level generator and text converter, with in-process parsers. `tools/generators-222/` is the Graal 2.22 terrain, dungeon and level generators as Python references. `formats/grv1-decoder` reads `Z3-V1.02` (11-bit codes) and `GR-V1.04` (a `HEIGHTS` block on a `GR-V1.03` body). `tools/level-tools-gaps/` writes the graal-txt and nw-txt file names and `bigmap.txt`, and the gif2mng frame rule on synthetic frames. DungeonMaker's crawler steps and the water, mountain and sand edge tiles stay undescribed. `tools/ghidra/` is the headless salvage pipeline (`analyze_tool.sh`, `ExportSalvageArtifacts.java`); raw decompiler output stays in gitignored `local-out/`. Trainer key loops are in `recovered-hacks/`.
- `tools/navigator/` — a local map of these notes (areas, the link web of one cluster, and the version eras) on the loopback address, port 8765. It does not phone home. See `tools/navigator/README.md`.
- `recovered-drive-tools/` and `recovered-hacks/` — behaviour notes on recovered original Graal tools and trainers: inventory, salvage placement, threat taxonomy, a defense checklist, and prose summaries. No binaries, decompiled C, binary addresses, or Drive file ids.
- `recovered-hacks/static-2026-11-01/` — static reads of AutoIt key macros (dig, idle, attack, water, random walk), a caps-lock utility, and the Win32 listserver's saved-login hand-off. No program was executed.
- `tools/level-convert/` — one converter across the level generations: `.zelda` (Z3-V1), `.graal` (GR-V1), `.nw` (GLEVNW01), a `.gmap` of level files, and a future `.seihai` package ([inf], not a format a 2.22 client reads).
- `archive/` — archival notes on decommissioned experimental walkers and tileset work (Ember / Avelan). Full assets live in a Drive zip; see `archive/ember-avelan-experimental.md`.
- `server-notes/public-allowed-versions.md` — which client versions public Reborn game servers allow: the two "Allowed:" lists on record (Graal the Adventure: 2.22 and 5.00r7; a gs2 server: 2.22 and a 1.41r1 to iPhone Belote range), how GServer-v2 reads `allowedversions.txt` ranges, and what that means for each 5.x and 6.x build.
- `server-notes/setcharprop-hash-n.md` — `#n` in a `setcharprop` value is the triggering player's nickname on script-running servers (not the NPC's); GServer13 has no `setcharprop`.

Newly scaffolded (2026-10-14)

These notes expand the existing foundation without adding raw decompile material:

- `formats/tile-walk.md` — sheet selection, type-0 allow lists, wall/water rules, stamp ids.
- `formats/sign-text.md` — network alphabet vs file alphabet, controls, drop-in decoder note.
- `formats/gobjset01.md` — stamp format and the 2×2 Bush example.
- `npc-library/gs1-commands.md` — control flow, common commands, pattern letters, offline wiring priority.
- `protocol/session-opcodes-cleaned.md` — common server-to-client meanings that match GLEVNW01 records.
- `client-internals/config-and-input.md` — graalconfig keys, device path, drop-files behavior.
- `client-internals/rendering-observations.md` — frame loop, draw path, offline blit next step.

- `client-internals/water-and-ganis.md` — the built-in animation name table, swim and lava, splashes, break-effect sounds (2.05).
- `client-internals/status-display.md` — the HUD: `state.png` rects, screen positions, heart rows and `showstats` bits (2.05 routine; 2.22 sheet and picture match).
- `client-internals/bare-level-names.md` — no client from 1st Graal to 2.05 adds `.zelda` to a bare level name or falls back between extensions; the editor adds `.graal` on save.
- `client-internals/id24-205.md` — id 24 in 2.05 is the same weapon-image list as later (tags 0 to 6, 60-call full-list countdown, UDP id 15 twin); corrects the "four-field form" note.
- `client-internals/damage-sources.md` — hearts taken by bombs, blasts, arrows and the other shots, thrown objects and baddies; hit boxes and the no-kill rule (2.05).
- `client-internals/combat-200-204.md` — the sword hit test and hurt packet in 2.00 to 2.04 compared with 2.05 rev 2 (2.04 is the same; what 2.00 to 2.03 lack, by build).
- `client-internals/status-bit-0x20.md` — STATUS bit 0x20 is the `hidesword` timer; how it is set and what it gates (2.05).
- `client-internals/sign-panel.md` — the sign panel from `letters.png` (glyph grid, frame, # button codes, paging keys) and what an NPC with no image draws (2.05).
- `client-internals/ledges-push-pull.md` — ledge jumps (array 8 per facing, the 8-tick curves), pushing and pulling stone blocks (array 16), the held grab, the futile pull, and the `pushed`/`pulled` trigger actions (2.05).
- `client-internals/live-trace-205.md` — live 2.05 traces: the client reports its own CURPOWER on a hit, ledge-jump timing and running in place, deep water, lava swim and the 101-tick burn, and the death/respawn sequence after a server CURPOWER 0 (respawn at the login point, CURPOWER 6, death loop).
- `client-internals/live-gserver13.md` — the real 1999 GServer13 with the 1.3 test client, live under Wine: server defaults and the new-account template, the login order (board in its own frame), walking and chat on the wire, bush regrow times, the Wine DirectInput keyboard fix for 1.x clients, and which builds start under Wine.
- `client-internals/player-mannerisms-live.md` — tick-by-tick player behaviour in real 2.05 rev 2 (memory + video) and 1.39 (video): walk, running in place and push timing, sliding along walls, lift/carry/hide (0.1 tile/tick)/throw, sword and bush cut, the spin attack (a second S press during a swing, seen online), hurt knockback and 40-tick blink, swim, grab/pull, the built-in gani timings across gani sets, the 2.05 GIF-only player images, and what Seihai poses differ.
- `client-internals/live-checks-editor-offline-sound.md` — live runs of 2.05, 2.22, 2.31 and GraalEditor: the Training button and where each client starts offline, the embedded tutorial beating a disk copy, `tiledefsOffline.txt` in every version, the 2.31 Editor button, Generate map, Edit flags, and engine sounds by action.
- `client-internals/body-colours-and-attributes.md` — how 2.05 colours a body (palette entries per slot, the 20 colours, new-player defaults), what the looks commands change, and GANI attribute sprites and `DEFAULTATTR` (the hat, props 37–41).
- `client-internals/board-respawn.md` — how cut bushes, lifted pots, signs and stones come back: GServer13 respawns any 2×2 change after 6–12 s (random 0.75–1.25 × `respawntime` 10, measured live), clients never regrow, 1.3 rolls bush/pot drops client-side and sends them peer to peer, 1.42+ drops are server-side; GServer-v2's rules for comparison.
- `client-internals/item-drops.md` — what bushes, pots and stones drop in every build: 1.x clients roll `Random(100)` (25 % green rupee, 2 % blue, 5 % heart, 5 % bombs) on any cut and on a lifted bush, a pot always gives a heart, 1.38 to 1.41 also send `PLI_ITEMADD`, and from NewMain 1.42 (not 2.00) the server drops; no client rolls arrows or drops for stones; the 2.05 floor item (life 164, pickup box, blink, bounce, `PLI_ITEMTAKE`/`ITEMDEL`), with lab captures of 1.39, 2.05 and 2.22.

- `client-internals/carry-hide-and-glove.md` — hiding under a carried bush or stone with D (mode 14, poses, sprites, STATUS 0x02), the glove scale (players start at 1, so pots lift) and lifting `canbecarried` NPCs (2.05).
- `client-internals/offline-training.md` — embedded tutorial levels vs session levels.
- `client-internals/client-assets.md` — pics1/picso sizes, tiledef line, levels/ counts.
- `client-internals/lab-captures-2026-10-25/` — real 2.22 lab captures: chests (open facing up only), packet 40 gates, CURPOWER units, respawn target, default weapon image ids.
- `client-internals/foot-test.md` — single-point movement test and wall/water predicates (the 2.05 client's real wall test is a box: see `client-internals/decomp-findings-2026-10-25-l/`).
- `client-internals/version-tokens.md` — public GNW token map and filename builds.
- `client-internals/build-catalogue.md` — every archived client build 1998–2001: hash, link date, token, encryption gen, list host, level formats.
- `client-internals/build-diff-204-205.md` — function-level diff of 2.04, 2.05 rev1 and 2.05 rev2 (NewWorld merge, VMT shift, ADP anti-debug).
- `client-internals/cross-version-timeline.md` — classes, GS1 commands and handled packet ids from 1.33 to the 2.1 tests.
- `version-profiles/matrix/` — the cross-era matrix, 1998 to 6.11 plus Graal3D: how each client thinks, acts and looks, a static survey of all 57 Delphi executables, live walking-speed runs of six eras, and Seihai's profiles compared. `matrix/rules-139.md` tags 1.39 death, hurt, knockback, sword reach and wall test (matrix unknown #1 for era C). `matrix/rules-1st-graal.md` tags the same five rules for 1st Graal; 1.32 is still open (Decompile folder re-listed 2026-11-01, no export). Outer sizes for 2.12–2.16, 2.3 and 3.x are recorded in the matrix as unmeasured, aside from the 3.04 Linux opening size already in the lab note.
- `client-internals/version-profiles/` — per-version client profiles 1.41–2.17 (TOML + resolver): wire ids sent and handled, props, login, files, and player-visible features, with a source tag on every value.
- `tools/` — index plus notes on the editor, shop, generators (`generators-222/` for the 2.22 programs), gif2mng, RemoteControl, and the 2001 NPC server (`npcserver-2001.md`).

Gap-fill pass (2026-10-21)

Sources: GServer-v2 and gs2-parser (GPLv3), gs2lib, graal-serverlist, OpenGS1 (MIT), the Graal Reborn 2.22 docs, and observed behavior of client 2.05 rev 2. The open-source refs are summarized and cited, not copied. Every new fact carries a tag: **[2.05]** confirmed in 2.05, **[srv]** server source only, **[doc]** 2.22 docs, **[cap]** captures, **[inf]** inferred.

- `protocol/codec-and-framing.md` — G-types, u16be+zlib frames, RAWDATA, encryption gens 1–6, the 2.05 gen-3 insertion.
- `protocol/packet-ids.md` — full PLO/PLI tables with 2.05 coverage (59 handled ids).
- `protocol/player-props.md` — PLPROP, NPCPROP, and BDPROP ids and encodings; 2.05 knows 0–42, 50, 51.
- `protocol/file-transfer.md`, `protocol/chat-pm-rc.md`, `protocol/listserver.md`, `protocol/udp-p2p.md`.
- `formats/level-binary.md`, `formats/gmap.md`.
- `npc-library/gs1-reference-205.md`, `client-internals/client-architecture-205.md`.
- Corrections, listed in CONSOLIDATED.md: login key, PLAYERPROPS movement, directions, the GR-V1 bit stream, BoardModify, no GS1 bytecode in 2.05.

Added 2026-10-21: editor, offline mode, sound

- `client-internals/level-editor-205.md`: the level editor built into the 2.05 client (keys, tile undo, object dialogs, save formats).
- `client-internals/offline-tutorial-205.md`: how 2.05 goes offline, where offline levels come from, and the 13 embedded tutorial levels.
- `client-internals/sound-and-music.md`: 2.05 effects, MIDI over MCI, and the Winamp hook, compared with BASS and voice chat in 2.17/2.22/2.31.
- `tools/graaleditor-vs-205.md`: GraalEditor 2.3.1.1 and GraalShop compared with the built-in editor.
- `formats/grv1-decoder/`: a small GR-V1 / Z3-V1 level decoder.

Cross-version pass (2026-10-21)

Static read of the 2.05, 2.17, 2.22, and 2.31 clients and `RemoteControl.exe` from the Graal Reborn 2.2.2.0 folder. Tag **[bin]**.

- `protocol/client-versions.md`: login type, token, cipher, and framing per client; client-side gen-5 send rules; server→client id sets per version; listserver logins (`GNW30123` mapped) and the `arrays2.dat` list config.
- `client-internals/feature-timeline.md`: release-note timeline 2.00–2.31, tied to the protocol changes.
- `tools/remotecontrol.md`: RC (type 6, `GSERV024`, gen 5) and NC (type 3, `NCL11012`) logins and id sets.

Version profiles

- `version-profiles/2.22-2.31.md` and `.toml`: per-version profiles for 2.22, 2.3 (inferred) and 2.31. Covers the wire protocol, the start window, server-list tabs, editor presence, options, GS1 symbol changes, and feature switches for Seihai.
- `version-profiles/3.04-3.153.md` and `.toml`: the same profile for the 3.x Linux 3.04 and Mac 3.153 clients (twins of 2.22 and 2.31): sent and handled ids, prop ranges, no tampercheck, listserver, 3D mode and other features. `version-profiles/lab-304-linux-2026-10-30.md` is the lab run of the real 3.04 client on GServer-v2 (login on the wire, no tampercheck, push delay, chat look, bush and lift sequences). `version-profiles/lab-mac-tiger-2026-10-31.md` is the lab run of the real Mac 4.20r6 and 3.153 clients on Mac OS X 10.4 in qemu (login lines, send ids, X2/Y2 walking, no flag sends).
- `version-profiles/4.04-4.20.toml`: the 4.x Windows 4.04r2, Mac 4.20r6 and Windows 4.20r8 clients in table form (token, server ids handled, client ids sent, login fields), from `protocol/v4x-clients.md` and `client-internals/live-behaviour-404.md`. Read by Seihai's `tools/gen_profile_fixture.py`.
- `version-profiles/lab-v5-2026-10-30.md`: the lab run of the real 5.00r7, 5.12 and 5.24r4 Windows clients on GServer-v2. It covers the Login-server mode that hides the player until S→C 178, movement and gani timings, push triggers, bush cut and respawn, carrying and hiding, the chat text look, and HUD differences. Stills are in `client-internals/mannerisms-shots/v5-*.png`.

Decompile findings, batch b (2026-10-24)

- `client-internals/decomp-findings-2026-10-24-b/`: tamper-check (PLI 39) derivation, 2.31 periodic traffic, payload layouts of server ids above 48 (2.17.1, 2.22, 2.31), the 2.00 client, open items, and a cross-check against the repo. Tables are in `version-profiles/` (`plo-layouts-217-231.tsv`, `plo-`

`handled-ids-by-build.tsv`, `pli-sent-ids-by-build-20x.tsv`, `tampercheck-words.tsv`, `version-tokens-by-build.tsv`, `drive-clients-exes.tsv`).
- `client-internals/drive-clients-inventory.md`: what the Drive clients set contains and which builds are still missing.

Decompile findings, batch c (2026-10-24)

- `client-internals/decomp-findings-2026-10-24-c/`: 20 builds from 2.00 to the 2.1 tests, NewWorld betas and NewMain 24/26 (Ghidra + IDR). Login tokens, handled server ids and sent client ids per build, handler changes by build, UDP channel, test builds, vocabulary, and corrections to batch b. Tables are in the same folder. The 2.1 test decompiles are lossy, so only their structure is reliable. Long-handler field layouts (PLAYERPROPS / OTHERPLPROPS / NPCPROPS / LEVELLINK) for 2.00–2.31, plus UDP ids 0/15/22 for 2.00–2.04 and the gen-3 insertion formula through 2.05, are in `version-profiles/plo-*-layouts.tsv`, `udp-ids-200-204.tsv` and `gen3-insertion-200-205.tsv`.

Decompile findings, batch d (2026-10-25)

- `client-internals/decomp-findings-2026-10-24-d/`: sword damage (attacker's sword power in half hearts, heal rule, hurt mode, the sword-power byte), the extra gates on a hit, and the respawn entry-save rule (gmap-list test, the 2.05-only name group) from the 2.05 rev 2, 2.17.1, 2.22 and 2.31 clients. Tables are in the same folder. Static read only, nothing was seen on a wire.

Decompile findings, batch e (2026-10-25)

- `client-internals/decomp-findings-2026-10-25-e/`: file-transfer caching rules (the three name lists, the window of four requests per kind, how replies settle names, when a local file counts as checked, where downloads are saved and with which time) from the 2.00, 2.05 rev 2, 2.17.1, 2.22 and 2.31 clients, checked against the 2.17, 2.22 and 2.31 captures already summarised in earlier notes. Tables are in the same folder. `04-name-index-case-link.md` is a later static read of 2.05 rev 2 and 2.22: duplicate names (newer file in 2.22, first hit in 2.05), the A–Z case fold, and the packet 40 link (a horse does not change it).

Decompile findings, batch f (2026-10-25)

- `client-internals/decomp-findings-2026-10-25-f/`: what real `GR-V1.03` levels show (two supplied files and an archive of 3113 binary levels): section shapes and the no-sign file ending, link arrivals written as `playerx`/`playery` plus an offset, the baddy section ending only at a first byte of 0x80 or more (x = 0 is a valid baddy), sign `#K(n)` escapes, chest sign index -1, and aggregate counts. Counts and structure only; no level content or file names. The table `corpus-stats.tsv` is in the same folder. Small corrections to the `grv1.py` decoder and `level-binary.md` are in a separate commit.

Level editor notes (2026-10-25)

- `tools/level-editor/`: the classic Graal level editor, with the editor built into the 2.22 client as the primary reference (`222-reference.md`) and every other build as a delta (`version-deltas.md`: 1st Graal to 2.31, the New World betas, the standalone GraalEditor). Also a cross-reference to the format notes and batch f (`formats-crossref.md`) and a Seihai tracking checklist (`seihai-tracking.md`, 2.22 feature list in implementation order, then per-version deltas). New: the text reader's `BOARD` rules, the 9 x 9 `HEIGHTS` block, save format by file extension, the warp-field defaults, the key map additions, change-log dated features, and a class timeline across every build. Mouse handling and two mode flags are structure-only.

Corrections: the tile id to sheet mapping in `client-internals/client-assets.md`, and the baddy section ending in `client-internals/level-editor-205.md`.

Decompile findings, batch g (2026-10-25)

- `client-internals/decomp-findings-2026-10-25-g/`: the three fields that were still unknown. Client id 24 (relayed as 32) is the sender's list of weapon-script images, and the one byte every real client sends is its "full list" marker with no images; the format, the sender's wrap timing, why 2.22 sends only with another player present, and the receiver's rules. Client id 46 is a content-free level-switch notice with a literal 1 (one send site, 2.31 only). Server id 178 is read with the clients' comma-text rules (white space splits unquoted items, so names with spaces must be quoted; the type prefix is dropped from the name). What a real list service sends for 178 is still unobserved. Tables are in the same folder. Static read plus source of the open-source servers; nothing new was seen on a wire.
- `client-internals/decomp-findings-2026-10-25-j/`: `Z3-V1` (`.zelda`) levels checked against 139 distinct real files (663 copies, 1st Graal to 15th Graal and the Java Zelda Online folder) and the Java, 1st Graal and 1.39 loaders. All files read to exactly their end. Findings: `Z3-V1.03` links carry decimals (124 of 207 have a decimal x, y, width or height) and negative arrivals mean "keep the player's coordinate" (151 links; 805 `GR-V1` links too); computer records are `x y type` plus a rest-of-line that is empty in `.03` and a three-verse text in `.04`, with `0x0A` allowed as a coordinate; the extension decides nothing (31 `Z3-V1.04` files sit under `.graal`, 694 `.zelda`-named files are `GR-V1`); every Graal client binary carries both magics. Counts and structure only; the tables `by-build.tsv` and `format-stats.tsv` are in the same folder.

Decompile findings, batch k (2026-10-25)

- `client-internals/decomp-findings-2026-10-25-k/`: the leftovers of the damage and file-transfer notes. A hit pushes the player one tile a tick for five ticks along a unit vector (zero means east), undoing a step when one of twelve body-box points is blocked (identical in 2.05 rev 2, 2.17.1, 2.22 and 2.31); packets 40 and 46 go through the sword's gates (AP 100, the shared once-a-second stamp taken before the guild test, the guild tag) with no attacker status test, power capped at one heart and 0 accepted; packet 46 is a box of strictly under 2 tiles around the centre (X+1.5, Y+2.0). The save-folder layouts per build: 1.33 to 1.41 save everything in one folder, 2.00 and 2.01 have no `hats`, both 2.1 test builds have no `maps`, and every build ignores a zero-length stream. Static read; nothing new was seen on a wire.

Decompile findings, 2026-10-28 (b)

- `client-internals/decomp-findings-2026-10-28-b/`: how 2.05 decides tile types. The default tileset has no type table, and each type test asks one `arrays.dat` list (array 2 throw-through, 4 chair, 5 and 6 bed, 1 water with lava included, 3 lava, 8 jumping stone per direction). Tilesets of the new kind use a fixed rule by sheet position. `tiletype()` returns -1 on the default tileset. Static read; `tile-type-lists.tsv` is in the same folder.

Precursors and fan builds (2026-10-29)

- `client-internals/precursors-and-fan-builds.md`: Zelda Online 1 and 2, Knorr's 1998 ZeldaEditor (whose dialogs became 1st Graal's editor), the GServ lobby with source, password saves, `objects.dat` and `arrays.dat`, Java Graal (2000), the Graal Game Boy Color demo (1999), 39ster's 2014 offline client, and how they relate.
- `client-internals/gameboy-demo.md`: the full disassembly of the Game Boy Color demo. It covers the GBDK/LCC build, the bank map, the room format (links, texts, NPC records) and all seven rooms, the 31

character types, controls, lifting, entity states and the sound player, and then what an emulator run showed: speeds, the sword, lift and throw, the per-character A items (bomb, boomerang), respawn, the room 4 event, and Start muting the game. The disassembly and the run's contact sheets are in the project's shared folder.
- `client-internals/precursor-runs-2026-10-30.md`: Zelda Online 2, Java Graal and the Game Boy Color demo run for real. It gives the game-feel numbers from the Java sources (20 Hz ticks, the 0.9 s push pose and 1.4 s stone push, the bush drop table, no respawn timer, chat as outlined text with no bubble) and what the wire showed (room names, a doubled relay when two clients pair at once).

Boundary

Raw decompiles stay out of this repository. These notes are derived from public server sources, observed packet shapes, format documentation, and observed client behavior. They contain no code or addresses from the binaries.

Version profiles 1.33–1.39 (2026-10-23)

- `client-internals/versions-133-139.md`: what changes between the seven gen-2 GNW clients, on the wire and for the player (HUD art, tiles, windows, controls, rules).
- `client-internals/version-profiles/v133-v139.toml`: the same as a machine-readable table, one `[[version]]` per build, with confidence notes and first-build feature flags.

Archive (2026-10-28)

- `archive/ember-avelan-experimental.md`: archival notes on the decommissioned Ember/Avelan experimental walkers, tileset variants, and the type-0 mask. Full project zip is on Google Drive under `/[Grok/[Projects/Graal Enigma Playerworld.zip`.

Game assets (2026-11-01)

- `archive/game-assets-2026-11-01.md`: inventory of the owner's Game Assets Drive folder. Sounds and graphics are RAR/7z packs (wav, mid, au, gani, png, gif, mng); levels in one opened dump are `GLEVNW01` and `GR-V1.03`; a world folder is a dated archive of levels, sometimes a gmap, with scripts in a separate pack; old sites are summarized by page title. Assets stay on Drive.
- `archive/game-assets-graphics-sound-2026-11-01.md`: the graphics and sound packs that inventory left as filenames. `heads.7z` is 34,369 files, mostly 32×560 head strips, plus a RAR5 copy of the same names and an older zip; `Graal File Pack.7z` is a `levels/`, `sounds/`, and `npcs/` tree; `MIDI.rar` is mid, mp3, and zip (no `.kar`). `pets.rar`, `backpals.rar`, and the small New World MIDI rar are opened in full. Assets stay on Drive.
- `archive/game-assets-worlds-2026-11-01.md`: the level dumps, Zelda rars, Zone gmaps, and a sample of the 87 archived server folders left closed by that inventory. Other than the alkaren baseline, `.nw` files are 330 `GLEVNW01` and 5 `GR-V1.03`. Dungeon gmaps are `GENERATED` plus one name; the opened terrain gmaps store a numeric `HEIGHTMAP` and do not use bare name rows.
- `archive/game-assets-websites-2026-11-01.md`: the eleven website archives that inventory left unread (forums mirror, Dragon site, guild vault, 2023 construction site, kingdoms library, NPC download site, forum SQL schema, 2019 developer site, flash intro, Chaos Continuum, ODiUM). Page purpose and sections only; no HTML, SQL, or shots committed.

Graal 4.x clients (2026-10-29)

- `protocol/v4x-clients.md`: Windows 4.20r8 (a third-party packet-logger build) and Mac OS X 4.20r6, read statically. The Delphi launcher plus C++ `GraalEngine.dll` split, `<login-host-4x>:14900` with no separate listserver protocol, tokens `G3D27076` / `G3D20126`, the login's PC-id fields, saved-password DES, gen 5 unchanged, 101 handled server ids (13 new over 3.153), the `GRPKG001` update packages, what the packet logger really logs, and the 4.04r2 Windows client unpacked from its Thinstall installer (token `G3D17026`, the `Graal3DEngine.dll` plug-in, 98 server ids). The 3D plug-in: Graal TCP plus Torque UDP on port + 1, the connect request (protocol 19, account, nickname and the password in clear), and the `BigData` events that carry Graal text inside Torque.

Reference reimplementations (2026-10-30)

- `client-internals/reference-reimplementations/`: xtjoeytx's 2dadventureclient and LevelScriptEditor weighed against the real-client notes, behaviour by behaviour (matches, invented, missing). Main finding: the `gr.x`/`gr.y` flags seen in lab captures come from GServer-v2's `-gr_movement` weapon in the default account, not from the clients themselves.

Trainer behaviour models (2026-11-01)

- `recovered-hacks/behaviour-models.md`: what the recovered trainers actually send. Keys, counts, and waits for the AutoIt macros (including Prayer and both water-can versions), Graal Load's 150 ms line loop, and the two AFK bots. Digger stays truncated. `recovered-hacks/reference_macros.py` replays those tables in-process.
- `recovered-hacks/remaining-2026-11-01.md`: the scroll bot's one-character shift, the text sender's missing key, the caps-lock stub left packed, and a second look at the truncated digger zip.

Graal 4.04r2 live behaviour (2026-10-30)

- `client-internals/live-behaviour-404.md`: the real 4.04r2 client under Wine against GServer-v2. Why the Login server never draws players (a server name starting with `login` turns every draw layer off) and how S→C 178 hands the client to the game server. Also the chat look (white bold text with a blue shadow, no bubble), the walk speed, the push delay, grab, sword, bush cut and lift as board modifies, server-driven respawn, carrying, hiding under the object with D, and throwing.

Classic server tools (2026-11-01)

- `tools/classic-tools/`: what the classic listserver ELF, the two stat-page builders, Graal Character Designer, GChar, GraalLevelGen and TextConvert read, show and never send. Small Python references parse the list frame, the 1999 account blob and a `CHR:1.00` line in-process. The Win32 listserver's saved-login hand-off is not reimplemented.
- `tools/generators-222/`: the three writers in the Graal 2.22 `generators` folder (terrain plasma, dungeon stamps, image level generator) as Python references. gif2mng, the editors, TextConvert and the character designers are described elsewhere and are not rebuilt here.

tools/release/how-to-read.md

How to read these notes

Graal Analysis 0.0.1 is a document set of the notes in this repository. The chapter PDFs follow the contents list in `README.md`. Chiaki tags `v0.0.1`.

Boundary

`README.md` and `AGENTS.md` in this volume state the boundary. The notes are cleaned observations. Raw decompiles stay out of the repository. A fact is written as plain behaviour and carries a source tag.

Source tags

`AGENTS.md` names these tags: `[2.05]`, `[bin]`, `[lab]`, `[cap]`, `[srv]`, `[doc]`, `[inf]`.

The gap-fill section of `README.md` glosses `[2.05]` as confirmed in 2.05, `[srv]` as server source only, `[doc]` as the 2.22 docs, `[cap]` as captures, and `[inf]` as inferred.

`version-profiles/matrix/README.md` (in the client-internals volume) glosses `[bin]` as read from the client binary, `[file]` as shipped files, `[live]` as seen in a running client, `[cap]` as a packet capture, `[srv]` as server source, `[doc]` as release notes, and `[inf]` as inferred.

Notes viewer

After `git pull` of the tagged source, run:

```
python3 tools/navigator/navigator.py
```

The viewer listens on loopback port **8765**. No virtualenv is required. `tools/navigator/README.md` describes the map UI. The program stays in the source tree and is not reprinted as a PDF.

Left in the source tree

NPC script bodies stay in `npc-library/` as `page.json` files. The script-name appendix lists each script's image name, genre, and topic.

Machine-readable `.toml` and `.tsv` tables stay next to the notes that describe them.

Drive links in three source notes are omitted from these PDFs.